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Dear Editor-in-Chief,

Please consider the manuscript, “Existing Multi-Material FFF Printers Are Sufficient For (Near) Arbitrary Coloration,” as a Research Paper for publication in *Computers & Graphics*.

LayerLoom is a computational fabrication workflow that uses ordinary 3MF geometry to encode layer-wise material schedules for multi-material fused filament fabrication. Instead of assigning each displayed color to a separate filament, LayerLoom converts colored regions into interleaved Z-bands assigned to the physical filaments already loaded in the printer. The resulting color-generating schedule travels with the model itself and can be processed by standard multi-material slicers as ordinary filament-assigned geometry.

The manuscript contributes a geometry-level representation for color scheduling, an algorithmic workflow for converting colored 3MF or GLB models into slicer-ready woven 3MF assemblies, and an experimental evaluation of the resulting printed color states. The validation includes printed gamut samples, calibrated CIE $L^*a^*b^*$ /CIEDE2000 image-based measurements, application-oriented models, slicer compatibility tests, and print-time/material-use benchmarks.

We believe the manuscript is well suited to *Computers & Graphics* because it frames color expansion in FFF as a graphics and computational-fabrication problem: how to encode appearance and material scheduling into a portable model representation rather than into custom hardware, custom G-code, or slicer-specific color-mixing tools. This is closely aligned with prior work in 3D printing, appearance reproduction, geometry processing, and fabrication-aware graphics workflows.

Reproducibility is supported through the manuscript, Supplementary Information, and an anonymized review archive. The software release, example 3MF files, analysis scripts, benchmark data, and figure-generation workflows will be archived with a persistent DOI before publication.

This manuscript has not been published and is not under consideration elsewhere.

Thank you for your consideration.

Sincerely,

Finn Beruldsen